# Dimensional reduction in supergravity and duality symmetries

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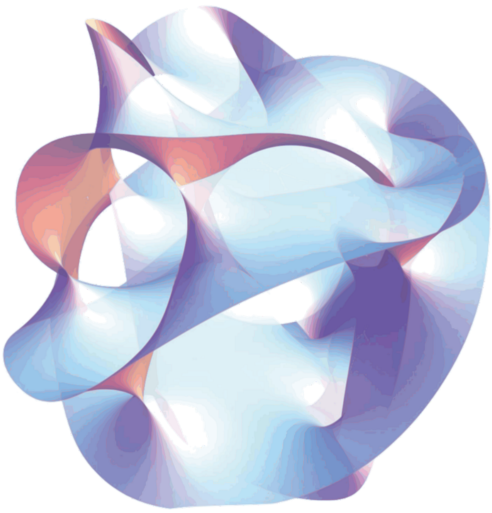

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## Acknowledgment

## Notations and abreviations

For the abreviations we wil use

- GR for general relativity

We will during this intership use the vielbein formalism. We will use the same notation as in [1] which is

- Greek hatted letter for curved indices $(\hat{\mu}, \hat{\nu}, \hat{\rho}, \dots)$
- Latin hatted letter for flat indices $(\hat{p}, \hat{m}, \hat{n}, \dots)$
- For space-time indices taking D values, late greek letters are used in the curved case $(\mu, \nu, \rho, \dots)$ and Latin letters in the flat case (p, m, n)
- For internal indices taking E values, early Greek letters in the curved case $(\alpha, \beta, \gamma, \dots)$ and Latin letters in the flat case (a,b,c)

THe general coordinate $x^{\hat{\mu}}$ will thereofre be made of space time coordinate $x^{\mu}$ and internal coordinate $y^{\alpha}$. The flat lorenz

# Contents

# 1 Introduction

Since the 19th-century unification of electricity and magnetism, and the later development of classical and quantum field theory, physicists have been looking for a single mathematical framework that could describe all fundamental interactions at once. A lot of progress has been made for the electromagnetic, weak and strong forces, which are now successfully unified in the Standard Model. Gravity, however, remains the main stumbling block. Unlike the other forces, it resists quantization in a straightforward way, and its description by General Relativity [2] sits completely apart from the quantum field-theoretic language we use for everything else.

One of the earliest attempts to change this was proposed by Kaluza in 1921 [3] and later refined by Klein:
by adding one extra spatial dimension compactified on a small circle, five-dimensional General Relativity naturally reduces to four-dimensional Einstein gravity coupled to electromagnetism. Even if the original theory had problems, this Kaluza-Klein mechanism opened a whole new way of thinking about unification, and it is now at the heart of modern approaches like supergravity [4] and string theory.

# 2 Theorical tool

Here you will find some GR reminder, which hopefully will make it easier to understand the mathematical foundation on which this internship was build

## 2.1 Groupe and gaude transformation

## 2.2 spin connexion, vielbein and flatspace

# 3 Lagrangian field theory

## 3.1 Lagrangian density

In field theory, just as Newton's second law is "volumized" into the Navier-Stokes equations by distributing quantities over space, we replace discrete degrees of freedom with continuous field densities defined at every point in spacetime. Therefore we will rewrite the Lagrangian as the integral over space of a new quantities called Lagrangian density, which will often refered to as just the Lagrangian. Thus we obtain this equation:

$$L = \iiint \mathcal{L}(\eta_\rho, \frac{\partial \eta_\rho}{\partial x^\nu})dxdydz \tag{1}$$

Now using esintein convention we can rewritte the action in a new light :

$$S = \int \mathcal{L} dx^\mu \tag{2}$$

## 3.2 General exemple for scalar and vectorial fields, and result for two forms

In field theory, the Lagrangian has a classical prescription for scalar and vector fields, but no general rule exists for other cases each field type requires its own treatment. For this work, only the scalar, vector, and two-form Lagrangians are needed, along with the metric, which is handled by Einstein's equations.
So for a scalar field we have :

$$\boxed{\mathcal{L} = -\frac{1}{2}\partial_\mu \Phi \partial^\mu \Phi} \tag{3}$$

and for a vectorial field we have

$$\boxed{\mathcal{L} = -\frac{1}{-}F_{\mu\nu}F^{\mu\nu}} \tag{4}$$

where for the field A

$$F_{\mu\nu} = \partial_\mu A^\nu - \partial_\nu A^\mu$$

In a vaccum the einstein equation give us

$$\boxed{\mathcal{L} = R}$$

where R is ricci scalar finally for the lagrangian of a two form fields whe will look in [5] where it is given by the equation :

$$\boxed{\mathcal{L} = \frac{1}{12} g^{\hat{\mu}\mu'} g^{\nu\nu'} g^{\rho\rho'} H_{\mu\nu\rho} H_{\mu'\nu'\rho'}} \tag{5}$$

where for the two form $B_{\nu\rho}$ we have

$$H_{\mu\nu\rho} = \partial_\mu B_{\nu\rho} + cyc.perm$$

# 4 Reduction of the NS-NS Action in D+E dimension

In this section we perform a Kaluza-Klein reduction. We consider a universe living in $D+E$ dimensions, with $D=4$, and we try to determine how an observer confined to the $D$-dimensional world perceives the extra dimensions.

## 4.1 Hypothesis

We assume that none of the fields depend on the internal coordinates, i.e. for any field $\partial_\alpha A^{\hat{\mu}} = 0$. This mean that the reduction will be done on a E-d torus.

## 4.2 Analysing the action

We start from the NS-NS action, which is

$$\boxed{S = \int dx^D \int \frac{dy^E}{\rho(E)} \sqrt{-\hat{g}} e^{\hat{\phi}} (\hat{R} + \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\nu}} \hat{\phi} - \frac{1}{12} g^{\hat{\mu}\mu'} g^{\nu\nu'} g^{\rho\rho'} H_{\mu\nu\rho} H_{\mu'\nu'\rho'})} \tag{6}$$

where $\rho(E)$ is a normalisation factor introduced in (6), corresponding in practice to the invariant volume of the internal space.
And where $\hat{R}$ is similar as the classical Hilbert-Einstein action but in $D+E$ dimension, $\hat{\phi}$ is a dilaton field and $-\frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho}$ is the action of a two form. Finally as no field depend on internal component and knowing what $\rho(E)$ we can just simplify this integral. Now we will separate this action into 3 part one for each action so we have

$$S_{\hat{g}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} \hat{R} \tag{7a}$$

$$S_{\hat{\phi}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} \hat{g}^{\hat{\mu}\hat{\nu}} \partial_{\hat{\mu}} \hat{\phi} \partial^{\hat{\nu}} \hat{\phi} \tag{7b}$$

$$S_{\hat{\phi}} = \int dx^D \sqrt{-\hat{g}} e^{-\hat{\phi}} - \frac{1}{12} H_{\mu\nu\rho} H^{\mu\nu\rho} \tag{7c}$$

Now using the fact that $\hat{R}$ can be written as

$$\hat{R} = \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}}$$

where

$$\hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}} = \partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{\omega}_{\hat{\mu}\hat{r}}{}^{\hat{t}} \omega_{\hat{\nu}\hat{t}\hat{s}} - (\mu \leftrightarrow \nu)$$

and the $\omega$ are defined By

$$\hat{\omega}_{trs} = -\hat{\Omega}_{\hat{t}\hat{r}\hat{s}} + \hat{\Omega}_{\hat{r}\hat{s}\hat{t}} - \hat{\Omega}_{\hat{s}\hat{t}\hat{r}}$$

and finally the $\Omega$ are defined thanks to the vielbein with

$$\hat{\Omega}_{\hat{r}\hat{s}\hat{t}} = \frac{1}{2} (\hat{V}_{\hat{r}}^{\hat{\mu}} \hat{V}_{\hat{s}}^{\hat{\nu}} - \hat{V}_{\hat{r}}^{\hat{\nu}} \hat{V}_{\hat{s}}^{\hat{\mu}}) \partial_{\hat{\nu}} \hat{V}_{\hat{\mu}\hat{r}}$$

with all of that we are going to rewrite the action $S_{\hat{g}}$ in termes of $\hat{\omega}$

$$\begin{aligned}
S_{\hat{g}} &= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{R} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \hat{R}_{\hat{\mu}\hat{\nu}\hat{r}\hat{s}} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} (\partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{\omega}_{\hat{\mu}\hat{r}}{}^{\hat{t}} \omega_{\hat{\nu}\hat{t}\hat{s}} - \partial_{\hat{\nu}} \hat{\omega}_{\hat{\mu}\hat{r}\hat{s}} - \hat{\omega}_{\hat{\nu}\hat{r}}{}^{\hat{t}} \omega_{\hat{\mu}\hat{t}\hat{s}}) \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} ([\partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} - \partial_{\hat{\nu}} \hat{\omega}_{\hat{\mu}\hat{r}\hat{s}}] + [\hat{\omega}_{\hat{\mu}\hat{r}}{}^{\hat{t}} \omega_{\hat{\nu}\hat{t}\hat{s}} - \hat{\omega}_{\hat{\nu}\hat{r}}{}^{\hat{t}} \omega_{\hat{\mu}\hat{t}\hat{s}}]) \\
&= \int dx^D \{2\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} + \hat{V} e^{-\hat{\phi}} [\hat{\omega}^{\hat{r}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{s}}{}_{\hat{t}\hat{s}} - \hat{\omega}^{\hat{s}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{r}}{}_{\hat{t}\hat{s}}]\} \\
&= \int dx^D \{\hat{V} e^{-\hat{\phi}} [\hat{\omega}^{\hat{r}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{s}}{}_{\hat{t}\hat{s}} - \hat{\omega}^{\hat{s}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{r}}{}_{\hat{t}\hat{s}}] + \underbrace{\partial_{\hat{\mu}} (2\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \hat{\omega}_{\hat{\nu}\hat{r}\hat{s}})}_{\text{Border terme which goes to zero}} - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} \partial_{\hat{\mu}} (\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}})\} \\
&= \int dx^D \{\hat{V} e^{-\hat{\phi}} [\hat{\omega}^{\hat{r}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{s}}{}_{\hat{t}\hat{s}} - \hat{\omega}^{\hat{s}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{r}}{}_{\hat{t}\hat{s}}] - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} \partial_{\hat{\mu}} (\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}})\} \\
&= \int dx^D \{\hat{V} e^{-\hat{\phi}} [\hat{\omega}^{\hat{r}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{s}}{}_{\hat{t}\hat{s}} - \hat{\omega}^{\hat{s}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{r}}{}_{\hat{t}\hat{s}}] - 2\hat{\omega}_{\hat{\nu}\hat{r}\hat{s}} [\hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \partial_{\hat{\mu}} \hat{V}^{\hat{\nu}\hat{s}} + \hat{V} e^{-\hat{\phi}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} \hat{V}^{\hat{\mu}\hat{r}} + e^{-\hat{\phi}} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} \hat{V} + \hat{V} \hat{V}^{\hat{\mu}\hat{r}} \hat{V}^{\hat{\nu}\hat{s}} \partial_{\hat{\mu}} e^{-\hat{\phi}}]\} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{[\hat{\omega}^{\hat{r}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{s}}{}_{\hat{t}\hat{s}} - \hat{\omega}^{\hat{s}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{r}}{}_{\hat{t}\hat{s}}] - 2\hat{\omega}_{\hat{\nu}}{}^{\hat{r}\hat{s}} [\hat{V}^{\hat{\mu}}_{\hat{r}} \hat{V}^{\hat{\nu}}_{\hat{s}} V^{\hat{\lambda}}_{\hat{t}} \partial_{\hat{\mu}} \hat{V}^{\hat{t}}_{\hat{\lambda}} - \hat{V}^{\hat{\mu}}_{\hat{r}} \hat{V}^{\hat{\nu}}_{\hat{t}} \hat{V}^{\hat{\lambda}}_{\hat{s}} \partial_{\hat{\mu}} \hat{V}^{\hat{t}}_{\hat{\lambda}} - \hat{V}^{\hat{\nu}}_{\hat{s}} \hat{V}^{\hat{\mu}}_{\hat{t}} \hat{V}^{\hat{\lambda}}_{\hat{r}} \partial_{\hat{\mu}} \hat{V}^{\hat{t}}_{\hat{\lambda}} - \hat{V}^{\hat{\mu}}_{\hat{r}} \hat{V}^{\hat{\nu}}_{\hat{s}} \partial_{\hat{\mu}} \hat{\phi}]\} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{[\hat{\omega}^{\hat{r}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{s}}{}_{\hat{t}\hat{s}} - \hat{\omega}^{\hat{s}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{r}}{}_{\hat{t}\hat{s}}] - 2\hat{\omega}_{\hat{s}}{}^{\hat{r}\hat{s}} [\hat{V}^{\hat{\mu}}_{\hat{r}} \hat{V}^{\hat{\lambda}}_{\hat{t}} \partial_{\hat{\mu}} \hat{V}^{\hat{t}}_{\hat{\lambda}} - \hat{V}^{\hat{\mu}}_{\hat{t}} \hat{V}^{\hat{\lambda}}_{\hat{r}} \partial_{\hat{\mu}} \hat{V}^{\hat{t}}_{\hat{\lambda}} - \hat{V}^{\hat{\mu}}_{\hat{r}} \partial_{\hat{\mu}} \hat{\phi}] + 2\hat{\omega}_{\hat{t}}{}^{\hat{r}\hat{s}} \hat{V}^{\hat{\mu}}_{\hat{r}} \hat{V}^{\hat{\lambda}}_{\hat{s}} \partial_{\hat{\mu}} \hat{V}^{\hat{t}}_{\hat{\lambda}}\} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{[\hat{\omega}^{\hat{r}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{s}}{}_{\hat{t}\hat{s}} - \hat{\omega}^{\hat{s}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{r}}{}_{\hat{t}\hat{s}}] - 2\hat{\omega}_{\hat{s}}{}^{\hat{r}\hat{s}} [2\hat{\Omega}_{\hat{t}\hat{r}}{}^{\hat{t}} - \hat{V}^{\hat{\mu}}_{\hat{r}} \partial_{\hat{\mu}} \hat{\phi}] + 2\hat{\omega}_{\hat{t}}{}^{\hat{r}\hat{s}} \hat{V}^{\hat{\mu}}_{\hat{r}} \hat{V}^{\hat{\lambda}}_{\hat{s}} \partial_{\hat{\mu}} \hat{V}^{\hat{t}}_{\hat{\lambda}}\} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{[\hat{\omega}^{\hat{r}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{s}}{}_{\hat{t}\hat{s}} - \hat{\omega}^{\hat{s}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{r}}{}_{\hat{t}\hat{s}}] + 2\hat{\omega}_{\hat{s}}{}^{\hat{r}\hat{s}} [2\hat{\omega}_{\hat{t}\hat{r}}{}^{\hat{t}} + \hat{V}^{\hat{\mu}}_{\hat{r}} \partial_{\hat{\mu}} \hat{\phi}] + 2\hat{\omega}_{\hat{t}}{}^{\hat{r}\hat{s}} \Omega_{\hat{r}\hat{s}}{}^{\hat{t}}\} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{[\hat{\omega}^{\hat{r}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{s}}{}_{\hat{t}\hat{s}} - \hat{\omega}^{\hat{s}}{}_{\hat{r}}{}^{\hat{t}} \omega^{\hat{r}}{}_{\hat{t}\hat{s}}] + 2\hat{\omega}_{\hat{s}}{}^{\hat{r}\hat{s}} [2\hat{\omega}_{\hat{t}\hat{r}}{}^{\hat{t}} + \hat{V}^{\hat{\mu}}_{\hat{r}} \partial_{\hat{\mu}} \hat{\phi}] - \hat{\omega}_{\hat{t}}{}^{\hat{r}\hat{s}} (\omega_{\hat{s}}{}^{\hat{t}}{}_{\hat{r}} + \hat{\omega}^{\hat{t}}{}_{\hat{r}\hat{s}})\} \\
&= \int dx^D \hat{V} e^{-\hat{\phi}} \{\hat{\omega}^{\hat{s}}{}_{\hat{s}}{}^{\hat{t}} \omega^{\hat{r}}{}_{\hat{r}\hat{t}} + \hat{\omega}^{\hat{s}\hat{t}}{}_{\hat{r}} \omega^{\hat{r}}{}_{\hat{s}\hat{t}} + 2\hat{\omega}_{\hat{s}}{}^{\hat{r}\hat{s}} \hat{V}^{\hat{\mu}}_{\hat{r}} \partial_{\hat{\mu}} \hat{\phi}\}
\end{aligned} \tag{8}$$

# 5 symmetries observation

# 6 Conclusion

# References


[1]J. Scherk and J. H. Schwarz, "How to get masses from extra dimensions", Nuclear Physics B **153**, 61–88 (1979).

[2]A. Einstein, "Zur allgemeinen relativitätstheorie", Sitzungsber. preuss. Akad. Wiss **44**, 778–786 (1915).

[3]T. Kaluza, "Zum unitätsproblem der physik", Sitzungsber. Preuss. Akad. Wiss. Berlin (Math. Phys.) **1921**, 966–972 (1921).

[4]D. Z. Freedman and A. Van Proeyen, *Supergravity* (Cambridge university press, 2012).

[5]J. Maharana and J. H. Schwarz, "Noncompact symmetries in string theory", Nuclear Physics B **390**, 3–32 (1993).


# A lagrangian

By using the principle of least action and the variational principle we can see that

$$\begin{aligned}
\delta S &= \delta \int \mathcal{L} dx^\mu \\
&= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho} \delta \eta_\rho + \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \partial_\nu \eta_\rho dx^\mu \\
&= \delta \int \frac{\partial \mathcal{L}}{\partial \eta_\rho} \delta \eta_\rho - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \eta_\rho + \underbrace{\partial_\nu \Big( \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} \delta \eta_\rho \Big)}_{\text{border terme equal to zero}} dx^\mu \\
&= 0
\end{aligned}$$

which gave the equation

$$\boxed{\frac{\partial \mathcal{L}}{\partial \eta_\rho} - \partial_\nu \frac{\partial \mathcal{L}}{\partial \partial_\nu \eta_\rho} = 0} \tag{9}$$

which is in fact the equation of mouvement for the field

# B General relativty 101

**Definition 1** *Let V be an R-vector space of dimension N, we define the tensor of order* $(r, s) \in [\![0, 1]\!]^2$ *on* $V^{(r,s)} \overset{def}{=} V^{\otimes r} \otimes V^{\otimes s}$ *as a being a multilinear application on* $V^{\star r} \times V^s$ *to* $\mathbb{R}$

**Definition 2** *We will call n-dimension* ($n \in \mathbb{N}^*$) *manifold every Hausdorff topological space* $(\mathcal{M}, \tau)$ *localy homeomorphic to* $\mathbb{R}^n$)

**Definition 3** *Let* $(\mathcal{M}, g)$ *be a smooth manifold. A metric tensor is a symmetric, non-degenerate bilinear form* $g_{\mu\nu}$ *on the tangent space* $T_p\mathcal{M}$ *at each point* $p \in \mathcal{M}$*, such that*

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

*In General Relativity we use the signature* $(-, +, +, +)$.

**Definition 4** *In general in RG we use the christofells connexion as it is the sole one wich is mtreci and torsion free. We define it as being*

$$nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu_{\mu\sigma} V^\sigma \tag{10}$$

*where the christofells symbols are*

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \tag{11}$$